

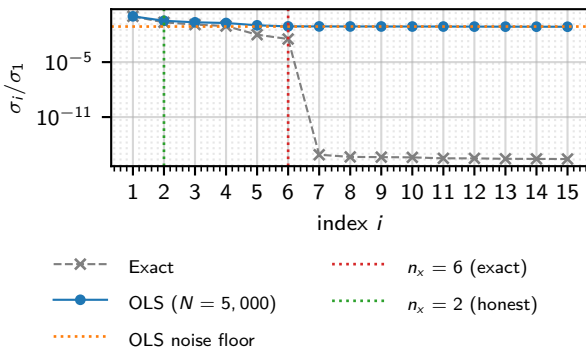
## Introduction

### System identification

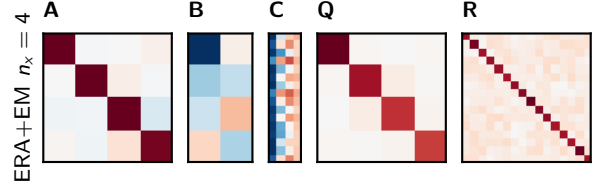
The exact Hankel singular-value spectrum, obtained via same-seed impulse subtraction (a theoretical reference not physically realisable with a real brain), has rank 6 (Figure 1). A physically realisable single-run OLS estimate ( $N = 5,000$  steps) raises a noise floor that obscures modes 3–6 (exact values  $\sigma_3/\sigma_1 = 0.125$  down to  $\sigma_6/\sigma_1 = 3.3 \times 10^{-3}$ ); BIC model selection recovers only  $n_x = 2$ . These modes carry little input–output energy and are near-unobservable from any finite honest recording. However, the downstream decoder operates on frequency-selective latent activity, so a model that misrepresents the transfer function will produce incorrect muscle activations regardless of one-step prediction quality. ERA+EM was therefore fitted at  $n_x \in \{2, 4, 6\}$  to identify the minimum sufficient order.

At  $n_x = 2$  the model reproduces per-neuron frequency selectivity for  $u_1$  but not for  $u_2$  (Figure 7). At  $n_x = 4$  both input channels are recovered;  $n_x = 6$  yields no visible further improvement. ERA+EM at  $n_x = 4$  was therefore selected. The Bode magnitude error (Figure 5), computed diagnostically via same-seed impulse subtraction, confirms this pattern: mean error falls substantially from  $n_x = 2$  to  $n_x = 4$ , with negligible reduction at  $n_x = 6$ .

CVA+EM was fitted at  $n_x = 4$  for comparison. Its transfer function reproduces the  $u_1$  selectivity but misrepresents  $u_2$  relative to the honest OLS estimate (rightmost column, Figure 7). One-step-ahead Kalman prediction on  $N = 100$  held-out trials ( $T = 2,000$  steps each) confirmed ERA+EM superiority across all output channels (Figure 6). ERA+EM at  $n_x = 4$  was adopted; the identified matrices are shown in Figure 2.



**Figure 1:** Normalised Hankel singular values  $\sigma_i/\sigma_1$  for the  $40 \times 60$  block Hankel matrix ( $n_u = 2$ ,  $n_y = 16$ ). The exact spectrum is obtained by same-seed impulse subtraction, which is not physically realisable, and serves as a theoretical reference for the true system order. Modes 3–6 fall below the OLS noise floor of a 5000-step physically realisable recording; BIC model selection on the OLS estimates recovers  $n_x = 2$ . [Source](#)



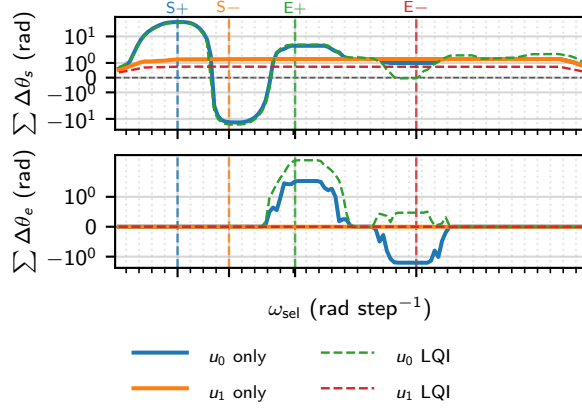
**Figure 2:** Identified system matrices for ERA+EM ( $n_x = 4$ ). Red indicates positive, blue negative; colour scales are symmetric about zero. [Source](#)

### Closed-loop spectral control

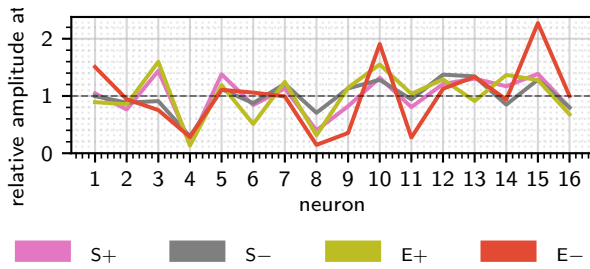
Having selected ERA+EM at  $n_x = 4$  as the working model, we test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population activity. The target is a sum of two sinusoids at  $f \in \{0.01, 0.03\}$  cycles per sample with amplitudes  $\{1.5, 0.7\}$ , oscillating about a physical population mean of 2.5 (well within the reachable range of  $[0.24, 5.4]$ ). Three controllers were implemented against this model: Linear–Quadratic–Gaussian (LQG), LQG with integral action (LQI), and Model Predictive Control (MPC).

All three use a Kalman filter for online state estimation. The LQR tracking cost is a per-neuron cost  $C^T C$  ( $\lambda_{\text{var}} = 1$ ), driving all 16 output channels simultaneously. LQI adds an integrator on the centred output error to reject steady-state bias; the integrator weight  $q_{\text{int}}$  is kept small so that sinusoidal tracking error is not accumulated. MPC solves a condensed receding-horizon quadratic programme ( $H = 30$  steps, covering one full period of  $f = 0.03$ ) with explicit input box constraints, warm-started between successive steps. Controller hyperparameters were selected via a log-uniform random search (8 trials per controller) scored on RMS tracking error.

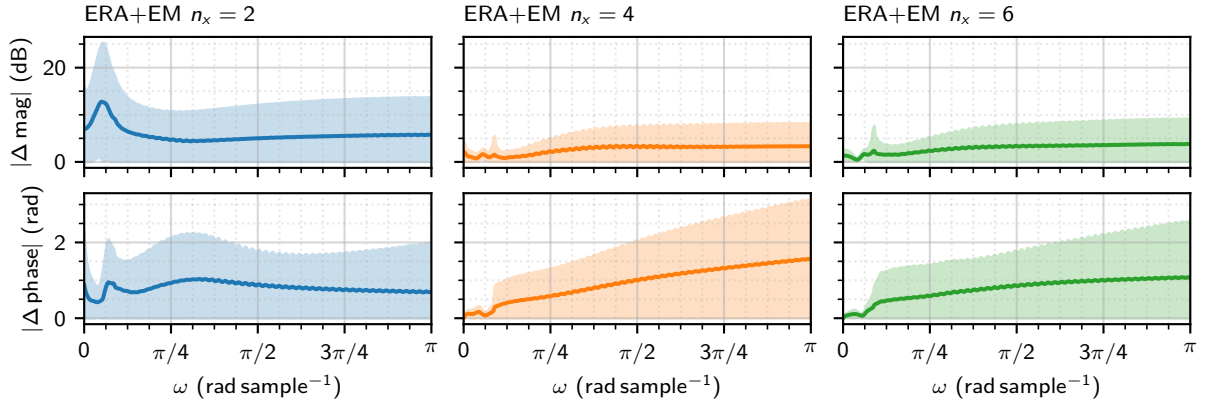
## 1 Muscle selection



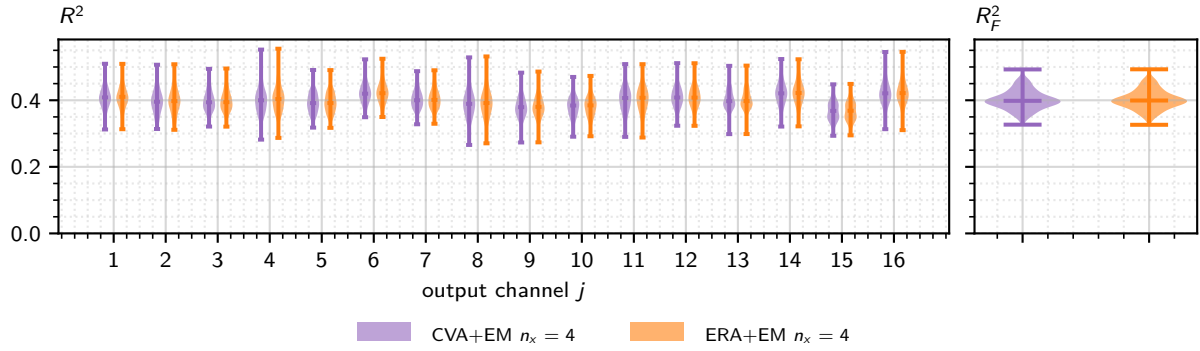
**Figure 3:** Muscle-selection frequency sweep. At each of 128 linearly spaced angular frequencies  $\omega_{\text{sel}} \in (0, \pi]$   $\text{rad step}^{-1}$ , four conditions are evaluated: open-loop sinusoidal drive (amplitude  $A = 0.5$ , centred at 0.5) applied to  $u_0$  or  $u_1$  independently (solid lines), and LQI closed-loop control targeting a population-mean oscillation of amplitude 1.0 through  $u_0$  or  $u_1$  alone (dashed lines). The arm is observed via hand position and inverse kinematics. Top: signed cumulative shoulder displacement  $\sum \Delta\theta_s$ ; bottom: cumulative elbow displacement  $\sum \Delta\theta_e$ , each summed over the post-warmup window of 800 steps (symlog y-axis, linear threshold  $\pm 1$  rad); curves smoothed with a 15-sample moving average. Coloured vertical lines mark the four empirically identified band frequencies (S+ at  $\omega^* = 0.417 \text{ rad step}^{-1}$ , S- at  $\omega^* = 0.761 \text{ rad step}^{-1}$ , E+ at  $\omega^* = 1.203 \text{ rad step}^{-1}$ , E- at  $\omega^* = 2.013 \text{ rad step}^{-1}$ ), identified from the open-loop  $u_0$  response by peak displacement magnitude (S $\pm$ ) and peak elbow selectivity  $|\Delta\theta_e|/(|\Delta\theta_e| + |\Delta\theta_s|)$  (E $\pm$ ). [Source](#)



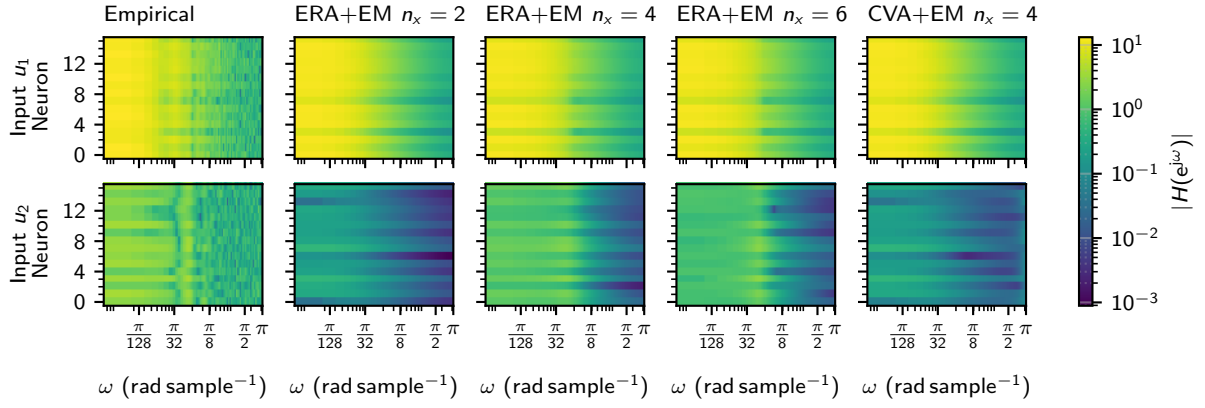
**Figure 4:** Neural correlates of muscle selection. For each of the four selection frequencies identified in [Figure 3](#), a sustained open-loop sinusoidal trial ( $u_0 = 0.5 + 0.4 \sin(2\pi f^* t)$ ,  $u_1 = 0.5$ ) recorded  $n_y = 16$  D neural observations over 1024 steps following 500 burn-in steps. Each line shows the amplitude at  $\omega_{\text{sel}}^*$  per neuron, normalised by the muscle's own neuron-mean; the dashed line marks the mean ( $= 1$ ). Profiles that collapse onto each other confirm that every muscle recruits the same relative neural activation pattern, so the  $16 \text{ D} \rightarrow x_1$  projection is not muscle-specific. [Source](#)



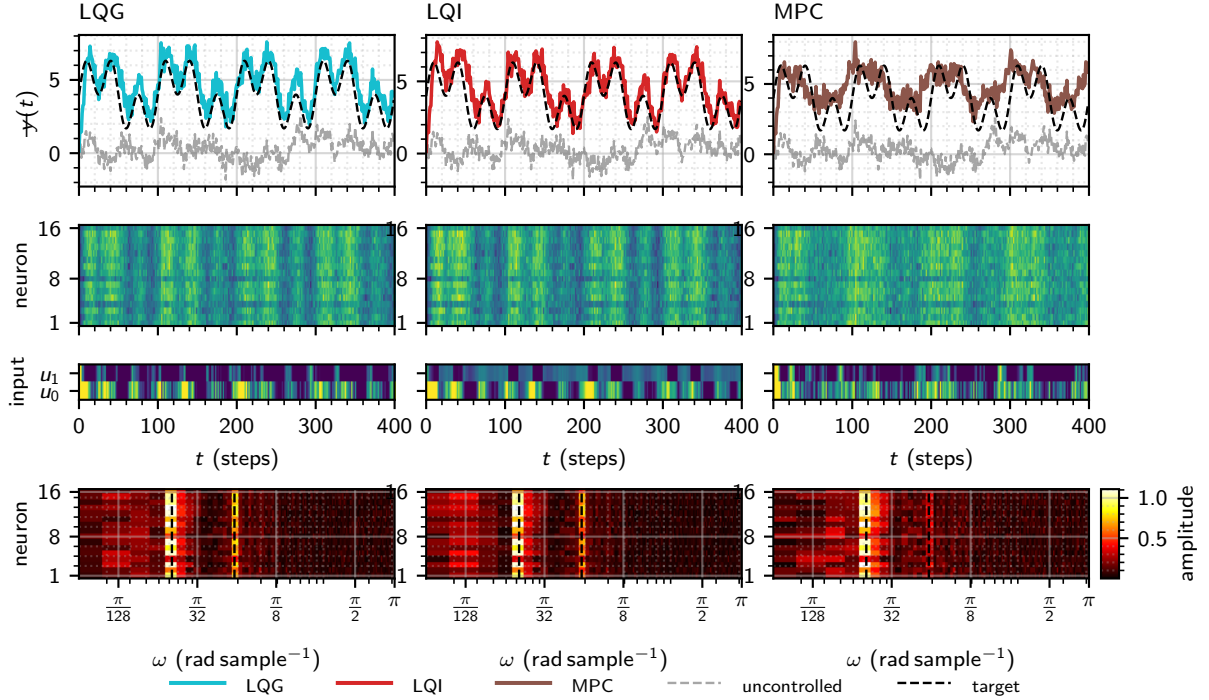
**Figure 5:** Magnitude error (top row) and phase error (bottom row) of ERA+EM transfer functions for  $n_x \in \{2, 4, 6\}$  (left to right). Each panel shows the absolute error between the exact empirical transfer function and the analytic model response  $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ , as mean  $\pm 1$  s.d. across all  $n_u n_y = 32$  output-input channels. The empirical reference is the exact discrete-time frequency response obtained via impulse subtraction (same-seed noise cancellation), DFT zero-padded to 512 points. Columns share a common x-axis; rows share a common y-axis. [Source](#)



**Figure 6:** Per-channel  $R^2$  (left,  $n_y = 16$ ) and Frobenius  $R_F^2$  (right) of one-step-ahead Kalman predictions for CVA+EM and ERA+EM at  $n_x = 4$ , across  $N = 100$  held-out trials of  $T = 2000$  steps (first 100 steps excluded as Kalman transient). Median: CVA+EM  $n_x = 4$   $R_F^2 = 0.398$ ; ERA+EM  $n_x = 4$   $R_F^2 = 0.399$ . [Source](#)



**Figure 7:** Per-neuron frequency selectivity for inputs  $u_1$  (top row) and  $u_2$  (bottom row). Each panel shows  $|H(e^{j\omega})|$  for all  $n_y = 16$  neurons (vertical axis) across  $\omega \in [0, \pi]$  rad sample $^{-1}$  (horizontal axis), on a shared logarithmic colour scale. Leftmost column: empirical response (honest OLS estimate from a single 5000-step recording, DFT zero-padded to 512 points). Columns 2–4: analytic ERA+EM transfer function  $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$  for  $n_x \in \{2, 4, 6\}$  respectively, showing the latent dimension sweep. Rightmost column: CVA+EM at  $n_x = 4$ , for estimator comparison at the chosen order. [Source](#)



**Figure 8:** Closed-loop spectral control using the ERA+EM  $n_x = 4$  model ( $n_x = 4$ ), controller hyperparameters tuned by random search (8 trials each). The target is a sum of two sinusoids at  $f \in \{0.01, 0.03\}$  cycles per sample with amplitudes  $A \in \{1.50, 1.50\}$ , centred on a physical mean of 4. Each column shows one controller (LQG, LQI, MPC) with a per-neuron tracking cost ( $\lambda_{\text{var}} = 1$ ). Row 1: physical population mean  $\bar{y}(t)$ , uncontrolled baseline (grey dashed), and target reference (black dashed). Row 2: per-neuron observations  $y(t)$  as a neuron  $\times$  time heatmap. Row 3: applied inputs  $u(t)$ . Row 4: per-neuron amplitude spectral density (neuron  $\times$  angular frequency  $\omega$ , log scale); target frequencies shown as red dashed lines. [Source](#)